

**T.C.
İZMİR UNIVERSITY OF ECONOMICS
FACULTY OF ARTS AND SCIENCES
DEPARTMENT OF MATHEMATICS**



LINEAR DISCRIMINANT ANALYSIS ON HUMAN DEVELOPMENT INDEX

Fatma Ebru ZOZUK

20220101024

Serhat BALIK

20200101003

MATHEMATICS THESIS ADVISER: Asst. Prof. Necla KOÇHAN

**MATH 497 FINAL THESIS/
MATH 497 MATHEMATICS THESIS**

İZMİR 2024

Abstract

Various frameworks have been created to assess and classify countries according to their level of development, and the Human Development Index (HDI) is the most logical to adopt widely. In the United Nations Development Programme publication, the HDI includes important development dimensions such as health, education, and income. Rather, perhaps the HDI's entrenchedness in research since its creation in 1990 is the subject of ongoing work, often concerned with their use and how to improve them to support more effective resource allocation and policy making. In this analysis, the sub-indices of the HDI – the Life Expectancy Index (LEI), the Education Index (EI), and the Income Index (II) – are analyzed using K-means clustering and Linear Discriminant Analysis (LDA). This R-based study is designed to cluster United Nations countries, and as a result, it will provide a more in-depth look at gaps in development and may help to better understand progress.

Keywords: Linear Discriminant Analysis, K-Means Clustering, Human Development Index, Dimensionality Reduction, Classification, Machine Learning, Statistical Analysis

1 Introduction

The Human Development Index (HDI) is a composite statistic of life expectancy, education, and income indicators, which are used to rank countries into tiers of human development. Introduced by the United Nations Development Programme (UNDP), the HDI provides a comprehensive measure of human well-being and development.

In such cases, data analysis techniques such as Linear Discriminant Analysis (LDA) and K-Means play a key role in understanding and classifying complex data sets. LDA projects high-dimensional data into a lower-dimensional space by maximizing class separation and minimizing intra-class variance. Applying LDA to HDI data allows us to group countries according to their development status and reveal underlying patterns in contributing variables.

In addition, the K-Means algorithm is a powerful clustering method used to explore the natural cluster structure of data and create meaningful groups in unlabeled data sets. By applying K-Means to HDI data, we can cluster countries with similar development levels and examine the relationships between variables. Using LDA and K-Means together helps us understand the complexity in HDI data by combining the advantages of both supervised and unsupervised learning approaches.

This study focuses on three path-related relationships (Life expectancy, years of education, and income level), which are the HDI dataset (Human Development Index) by using LDA (Linear Discriminant Analysis) and K-Means algorithms. In the case of LDA, it is a supervised learning method in which the countries are classified in terms of development according to their development levels, whereas in the case of K-Means, it is an unsupervised learning method to find the natural clusters of countries with similar characteristics. These two techniques are used simultaneously and it allows to understand how the countries manifest human development and the primary differences between the groups. The study aims to ease the decision-making process of policymakers by revealing the development distribution in the world.

The Human Development Index (HDI) has been extensively studied as a composite measure of national development, incorporating dimensions of health, education, and income. Researchers have applied various statistical and machine learning techniques to analyze and classify HDI data, aiming to enhance its interpretability and utility for policymakers.

Mylevaganam (2017) conducted a comprehensive analysis of HDI to categorize United Nations (UN) member states into development levels, emphasizing the role of tailored analytical frameworks for improved classification accuracy.

Regionally, Fahmiyah and Ningrum (2023) applied K-means clustering to HDI data in Indonesia, revealing regional disparities and demonstrating the efficacy of clustering in uncovering hidden patterns within development indicators. Their findings underscored the value of localized analytical approaches in addressing inequities.

Rocha et al. (2021) focused on the computational aspects of HDI analysis, exploring the application of K-means and related clustering techniques for approximating HDI values. Their work highlighted the advantages of algorithmic approaches in handling complex, multidimensional datasets, enabling more precise classification and understanding of HDI trends.

From a statistical perspective, Burmaoğlu and Oktay (2015) employed discriminant analysis to classify countries based on their HDI levels. Their study demonstrated the effectiveness of discriminant analysis in predicting group membership, providing a robust statistical framework for analyzing development disparities. Similarly, Iwuagwu and Nwosu (2020) applied discriminant analysis to model HDI, offering insights into the predictive capacity of this technique in understanding and categorizing development indicators.

Collectively, these studies illustrate a growing trend toward integrating advanced statistical and computational techniques for HDI analysis. The application of methods such as K-means clustering and discriminant analysis not only enhances the precision of HDI classifications but also fosters a deeper understanding of global and regional developmental patterns. Such insights are instrumental for data-driven policy interventions aimed at reducing inequalities and promoting sustainable development.¹

¹Supervised and unsupervised learning methods are two fundamental approaches in machine learning. Supervised learning works with labeled data and aims to train a model to predict the correct output for given inputs (e.g., classifying emails as spam or not). In contrast, unsupervised learning uses unlabeled data to discover hidden patterns or structures, such as clustering customers based on purchasing behavior.

2 Methodology

2.1 Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction technique widely used for classification problems. The primary goal of LDA is to project high-dimensional data onto a lower-dimensional space in a way that maximizes the separability of classes. This section provides a comprehensive explanation of the mathematical foundation of LDA.

2.1.1 Mathematical Foundation

The key idea of LDA is to find a projection direction, $\mathbf{w}$, that maximizes the ratio of the between-class variance to the within-class variance [2]. This is expressed as the Fisher Index, defined mathematically as:

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}, \quad (1)$$

where:

- $\mathbf{W}^T$: Transpose of the Matrix.
- $\mathbf{S}_B$: Between-class scatter matrix.
- $\mathbf{S}_W$: Within-class scatter matrix.
- $\mathbf{w}$: Projection vector.

2.1.1.1 Between-Class Scatter Matrix The between-class scatter matrix $\mathbf{S}_B$ captures the spread between the mean vectors of different classes and is calculated as:

$$\mathbf{S} = \sum_{i=1}^N N_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^T$$

- $\mathbf{S}$: Scatter matrix (e.g., between-class scatter matrix).
- N_i : Number of samples in class i .
- $\boldsymbol{\mu}_i$: Mean vector of class i .
- $\boldsymbol{\mu}$: Overall mean vector (mean of all classes).
- $(\boldsymbol{\mu}_i - \boldsymbol{\mu})$: Deviation of class i 's mean from the overall mean.
- $(\boldsymbol{\mu}_i - \boldsymbol{\mu})^T$: Transpose of the deviation vector.

Within-Class Scatter Matrix The within-class scatter matrix $\mathbf{S}_W$ captures the spread of samples within each class and is defined as:

$$\mathbf{S}_W = \sum_{i=1}^C \sum_{\mathbf{x} \in \mathcal{X}_i} (\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^T$$

- S_W : The **within-class scatter matrix**, which measures the scatter of samples within each class.
- $\sum_{i=1}^C$: Summation over all classes, from $i = 1$ to C (number of classes).
- $\sum_{x \in \mathcal{X}_i}$: Summation over all data points x belonging to class i .
- $\mathbf{x}$: A data point in the dataset.
- $\boldsymbol{\mu}_i$: The mean vector of class i .
- $(\mathbf{x} - \boldsymbol{\mu}_i)$: The deviation of a data point from the mean of its class.
- $(\mathbf{x} - \boldsymbol{\mu}_i)^T$: The transpose of the deviation vector.

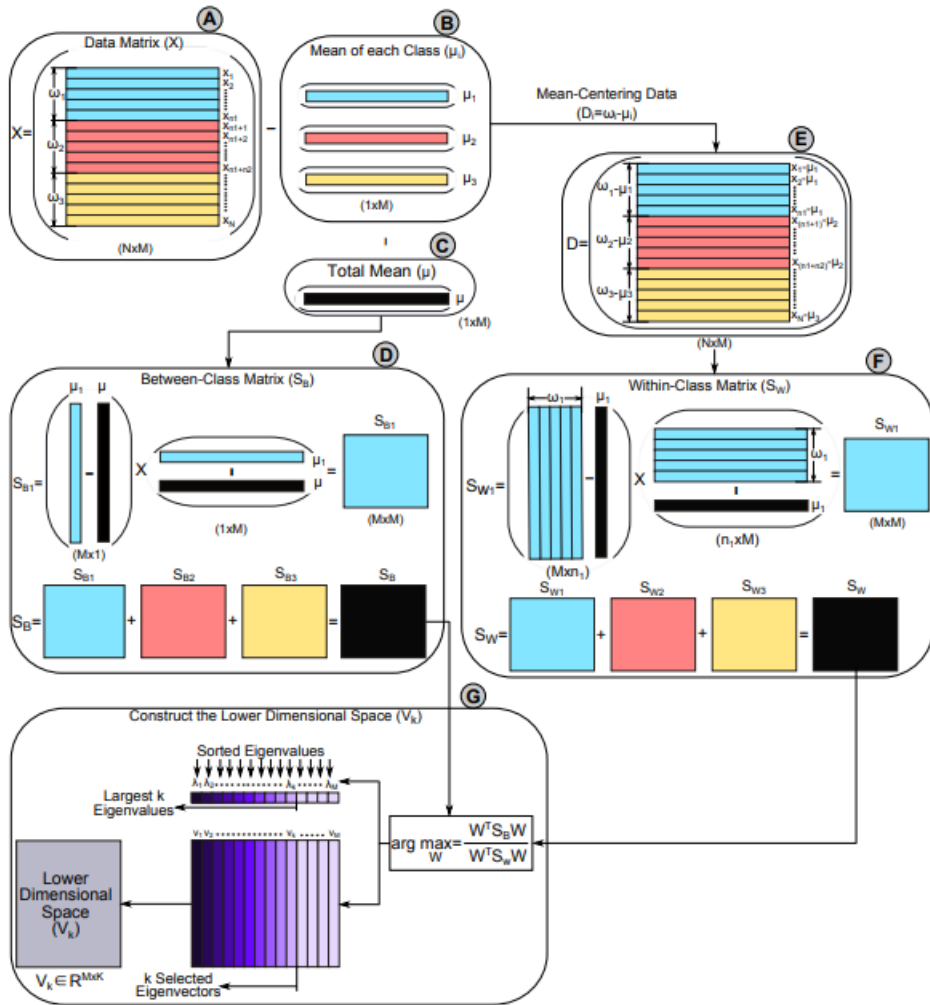


Figure 1: Visualized steps to calculate a lower dimensional subspace of the LDA technique.

2.1.2 Optimization Procedure

To maximize $J(\mathbf{w})$, the optimal projection vector $\mathbf{w}$ is obtained by solving the eigenvalue problem:

$$\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w},$$

where λ is the eigenvalue corresponding to $\mathbf{w}$.

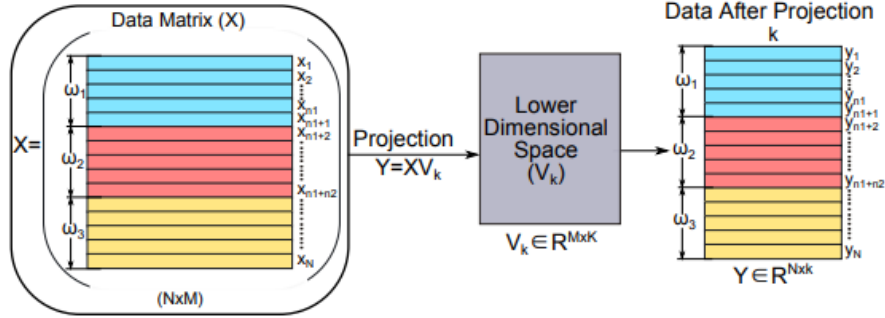


Figure 2: Projection of the original samples (i.e. data matrix) on the lower dimensional space of LDA (V_k)

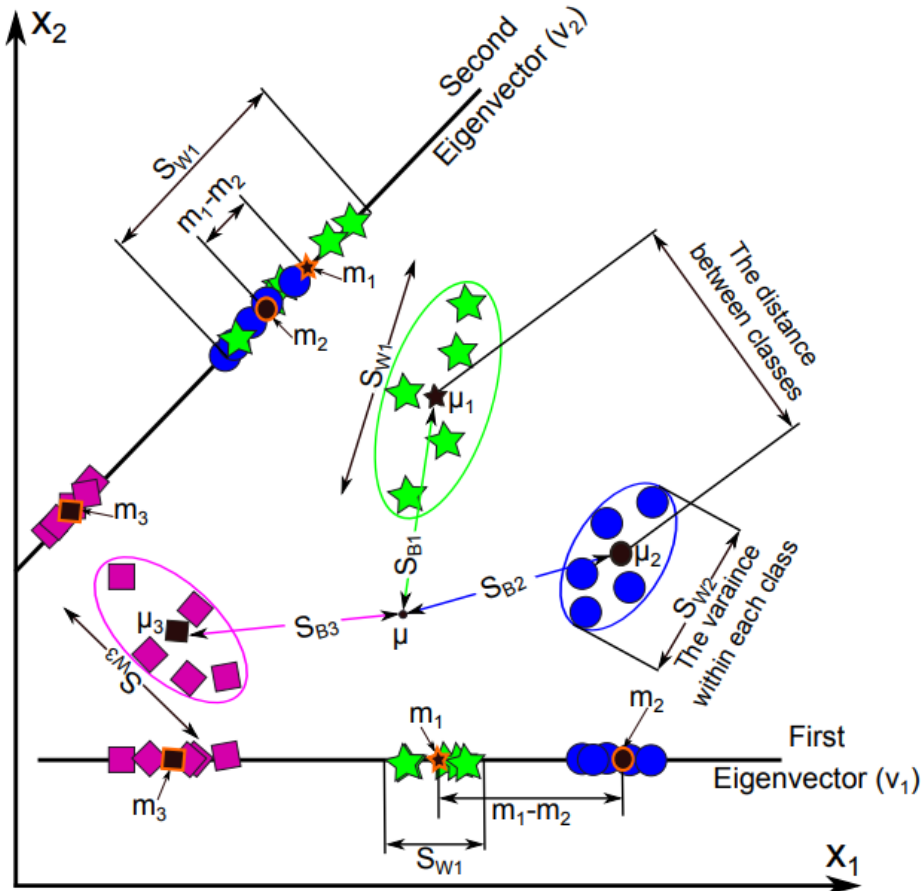


Figure 3: A visualized comparison between the two lower-dimensional sub-spaces which are calculated using three different classes.

Notation	Description	Notation	Description
X	Data matrix	x_i	i^{th} sample
N	Total number of samples in X	M	Dimension of X or the number of features of X
W	Transformation matrix	V_k	The lower dimensional space
n_i	Number of samples in ω_i	c	Total number of classes
μ_i	The mean of the i^{th} class	μ'_i	The mean of the i^{th} class after projection
μ	Total or global mean of all samples	m	The total mean of all classes after projection
S_{W_i}	Within-class variance or scatter matrix of the i^{th} class (ω_i)	S_W	Within-class variance
S_{B_i}	Between-class variance of the i^{th} class (ω_i)	S_B	Between-class variance
V	Eigenvectors of W	λ	Eigenvalue matrix
v_i	i^{th} eigenvector	λ_i	i^{th} eigenvalue
x_{ij}	The i^{th} sample in the j^{th} class	Y	Projection of the original data
k	The dimension of the lower dimensional space (V_k)	ω_i	i^{th} class

Table 1: Notation and Description Table for Linear Discriminant Analysis

2.2 K-Means Algorithm

K-means algorithm is a clustering algorithm that partitions data into k distinct clusters. The objective is to group similar data points together and minimize the sum of squared errors within each cluster.

2.2.1 Objective Function (Sum of Squared Errors)

The objective function J is defined as:

$$J = \sum_{i=1}^k \sum_{x_j \in C_i} \|x_j - \mu_i\|^2, \quad (3)$$

where:

- k : Number of clusters.
- C_i : The i -th cluster.
- x_j : The j -th data point.
- μ_i : The centroid of the i -th cluster.
- $\|x_j - \mu_i\|^2$: The squared Euclidean distance between the data point and the centroid.

2.2.2 Algorithm Steps

1. **Initialization Step:** Randomly select k initial centroids $(\mu_1, \mu_2, \dots, \mu_k)$.
2. **Assignment Step:** Assign each data point x_j to the nearest centroid μ_i :

$$C_i = \{x_j : \|x_j - \mu_i\|^2 \leq \|x_j - \mu_l\|^2, \forall l \neq i\}. \quad (4)$$

3. **Update Step:** Recalculate the centroids by taking the mean of all points assigned to each cluster:

$$\mu_i = \frac{1}{|C_i|} \sum_{x_j \in C_i} x_j. \quad (5)$$

4. **Iteration:** Repeat the assignment and update steps until the centroids do not change significantly or a specified number of iterations is reached.

Pseudocode for K-Means Algorithm

Input: Dataset X , number of clusters k

Output: k clusters with centroids

1. Initialize k centroids randomly.
2. Repeat until convergence:
 - a. For each data point x_j :
Assign x_j to the nearest centroid (cluster).
 - b. For each cluster C_i :
Update the centroid μ_i as the mean of points in C_i .
3. Return the final clusters and centroids.

Different Distance Metrics and Their Impact

In the K-means algorithm, the clustering result depends significantly on the choice of distance metric. The most common metrics include:

1. Euclidean Distance:

$$\|x - y\| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}. \quad (6)$$

This metric works well when clusters are spherical and the dimensions are equally scaled.

2. Manhattan Distance:

$$\|x - y\|_1 = \sum_{i=1}^n |x_i - y_i|. \quad (7)$$

This is useful when dealing with grid-like or high-dimensional data.

3. Minkowski Distance:

$$\|x - y\|_p = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}. \quad (8)$$

When $p = 2$, this is equivalent to the Euclidean distance. For $p = 1$, it becomes the Manhattan distance.

4. Cosine Distance:

$$d_{\text{cosine}} = 1 - \frac{x \cdot y}{\|x\| \|y\|}. \quad (9)$$

This is suitable for text data or when the magnitude of vectors is less important than their direction.

2.3 Impact of Different Distances

Choosing different distance metrics can lead to different clustering results:

- **Euclidean Distance** tends to create circular clusters.
- **Manhattan Distance** creates diamond-shaped clusters.
- **Cosine Distance** captures angular relationships, which can be useful in text mining.

Selecting the appropriate distance metric depends on the nature of the dataset and the problem context.

3 Modeling Sample Questions

Calculations for Class 1

The dataset for this example is shown below:

X_1	X_2	Class
4	1	1
2	4	1
2	3	1
3	6	1
4	4	1

Table 2: Dataset for Class 1.

3.0.1 Mean Matrix

The mean matrix is defined as:

$$\mu = [\mu_{X_1}, \mu_{X_2}] \quad (10)$$

where:

- μ_{X_1} = Mean of Feature 1 (X_1) for Class 1
- μ_{X_2} = Mean of Feature 2 (X_2) for Class 1

From the dataset, the means are calculated as follows:

$$\mu_{X_1} = \frac{4 + 2 + 2 + 3 + 4}{5} = 3 \quad (11)$$

$$\mu_{X_2} = \frac{1 + 4 + 3 + 6 + 4}{5} = 3.6 \quad (12)$$

Thus, the mean matrix is:

$$\mu = [3, 3.6] \quad (13)$$

3.0.2 Covariance Matrix

The covariance matrix measures the joint variability of two variables. It is calculated using the formula:

$$S_k = (X_k - \mu)(X_k - \mu)^T \quad (14)$$

where:

- X_k is the data point (features X_1 and X_2).
- μ is the mean vector of X_1 and X_2 .

3.0.3 Step-by-Step Covariance Calculation

The differences $X_1 - \mu_1$ and $X_2 - \mu_2$ for each row are calculated first, as shown below:

X_1	X_2	$X_1 - \mu_1$	$X_2 - \mu_2$
4	1	1	-2.6
2	4	-1	0.4
2	3	-1	-0.6
3	6	0	2.4
4	4	1	0.4

Table 3: Differences from the mean for Class 1.

3.0.4 Covariance Matrices (S_1 to S_5)

3.0.4.1 S_1 : For the first data point:

$$X_1 - \mu = \begin{bmatrix} 1 \\ -2.6 \end{bmatrix}$$

$$S_1 = (X_1 - \mu)(X_1 - \mu)^T = \begin{bmatrix} 1 \\ -2.6 \end{bmatrix} \begin{bmatrix} 1 & -2.6 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & -2.6 \\ -2.6 & 6.76 \end{bmatrix}$$

3.0.4.2 S_2 : For the second data point:

$$X_2 - \mu = \begin{bmatrix} -1 \\ 0.4 \end{bmatrix}$$

$$S_2 = (X_2 - \mu)(X_2 - \mu)^T = \begin{bmatrix} -1 \\ 0.4 \end{bmatrix} \begin{bmatrix} -1 & 0.4 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 1 & -0.4 \\ -0.4 & 0.16 \end{bmatrix}$$

3.0.4.3 S_3 : For the third data point:

$$X_3 - \mu = \begin{bmatrix} -1 \\ -0.6 \end{bmatrix}$$

$$S_3 = (X_3 - \mu)(X_3 - \mu)^T = \begin{bmatrix} -1 \\ -0.6 \end{bmatrix} \begin{bmatrix} -1 & -0.6 \end{bmatrix}$$

$$S_3 = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 0.36 \end{bmatrix}$$

3.0.4.4 S_4 : For the fourth data point:

$$X_4 - \mu = \begin{bmatrix} 0 \\ 2.4 \end{bmatrix}$$

$$S_4 = (X_4 - \mu)(X_4 - \mu)^T = \begin{bmatrix} 0 \\ 2.4 \end{bmatrix} \begin{bmatrix} 0 & 2.4 \end{bmatrix}$$

$$S_4 = \begin{bmatrix} 0 & 0 \\ 0 & 5.76 \end{bmatrix}$$

3.0.4.5 S_5 : For the fifth data point:

$$X_5 - \mu = \begin{bmatrix} 1 \\ 0.4 \end{bmatrix}$$

$$S_5 = (X_5 - \mu)(X_5 - \mu)^T = \begin{bmatrix} 1 \\ 0.4 \end{bmatrix} \begin{bmatrix} 1 & 0.4 \end{bmatrix}$$

$$S_5 = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 0.16 \end{bmatrix}$$

Summing the Covariance Matrices The total covariance matrix for Class 1 is:

$$S_{C1} = \frac{1}{n} \sum_{k=1}^n S_k$$

$$S_{C1} = \frac{1}{5} \left(\begin{bmatrix} 1 & -2.6 \\ -2.6 & 6.76 \end{bmatrix} + \begin{bmatrix} 1 & -0.4 \\ -0.4 & 0.16 \end{bmatrix} + \begin{bmatrix} 1 & 0.6 \\ 0.6 & 0.36 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 5.76 \end{bmatrix} + \begin{bmatrix} 1 & 0.4 \\ 0.4 & 0.16 \end{bmatrix} \right)$$

$$S_{C1} = \begin{bmatrix} 0.8 & -0.4 \\ -0.4 & 2.64 \end{bmatrix}$$

Calculations for Class 2

Step 1: Class Data and Mean Matrix

X_1	X_2	Class
9	10	2
6	8	2
9	5	2
8	7	2
10	8	2

Table 4: Dataset for Class 2.

3.0.5 Mean Matrix

The mean matrix is defined as:

$$\mu = [\mu_{X_1}, \mu_{X_2}] \tag{15}$$

where:

- μ_{X_1} = Mean of Feature 1 (X_1) for Class 2
- μ_{X_2} = Mean of Feature 2 (X_2) for Class 2

From the dataset, the means are calculated as follows:

$$\mu_{X_1} = \frac{9 + 6 + 9 + 8 + 10}{5} = 8.4 \quad (16)$$

$$\mu_{X_2} = \frac{10 + 8 + 5 + 7 + 8}{5} = 7.6 \quad (17)$$

Thus, the mean matrix is:

$$\mu = [8.4, 7.6] \quad (18)$$

3.0.6 Covariance Matrix

3.0.7 Step-by-Step Covariance Calculation

The differences $X_1 - \mu_1$ and $X_2 - \mu_2$ for each row are calculated first, as shown below:

X_1	X_2	$X_1 - \mu_1$	$X_2 - \mu_2$
9	10	0.6	-2.4
6	8	-2.4	0.4
9	5	0.6	-2.6
8	7	-0.4	-0.6
10	8	1.6	0.4

Table 5: Differences from the mean for Class 2.

3.0.8 Covariance Matrices (S_6 to S_{10})

3.0.8.1 S_6 : For the sixth data point:

$$X_1 - \mu = \begin{bmatrix} 0.6 \\ -2.4 \end{bmatrix}$$

$$S_6 = (X_1 - \mu)(X_1 - \mu)^T = \begin{bmatrix} 0.6 \\ -2.4 \end{bmatrix} \begin{bmatrix} 0.6 & -2.4 \end{bmatrix}$$

$$S_6 = \begin{bmatrix} 0.36 & -1.44 \\ -1.44 & 5.74 \end{bmatrix}$$

3.0.8.2 S_7 : For the seventh data point:

$$X_2 - \mu = \begin{bmatrix} -2.4 \\ 0.4 \end{bmatrix}$$

$$S_7 = (X_2 - \mu)(X_2 - \mu)^T = \begin{bmatrix} -2.4 \\ 0.4 \end{bmatrix} \begin{bmatrix} -2.4 & 0.4 \end{bmatrix}$$

$$S_7 = \begin{bmatrix} 5.76 & -0.96 \\ -0.96 & 0.16 \end{bmatrix}$$

3.0.8.3 S_8 : For the eighth data point:

$$X_3 - \mu = \begin{bmatrix} 0.6 \\ -2.6 \end{bmatrix}$$

$$S_8 = (X_3 - \mu)(X_3 - \mu)^T = \begin{bmatrix} 0.6 \\ -2.6 \end{bmatrix} \begin{bmatrix} 0.6 & -2.6 \end{bmatrix}$$

$$S_8 = \begin{bmatrix} 0.36 & -1.56 \\ -1.56 & 6.76 \end{bmatrix}$$

3.0.8.4 S_9 : For the ninth data point:

$$X_4 - \mu = \begin{bmatrix} -0.4 \\ -0.6 \end{bmatrix}$$

$$S_9 = (X_4 - \mu)(X_4 - \mu)^T = \begin{bmatrix} -0.4 \\ -0.6 \end{bmatrix} \begin{bmatrix} -0.4 & -0.6 \end{bmatrix}$$

$$S_9 = \begin{bmatrix} 0.16 & 0.24 \\ 0.24 & 0.36 \end{bmatrix}$$

3.0.8.5 S_{10} : For the tenth data point:

$$X_5 - \mu = \begin{bmatrix} 1.6 \\ 0.4 \end{bmatrix}$$

$$S_{10} = (X_5 - \mu)(X_5 - \mu)^T = \begin{bmatrix} 1.6 \\ 0.4 \end{bmatrix} \begin{bmatrix} 1.6 & 0.4 \end{bmatrix}$$

$$S_{10} = \begin{bmatrix} 2.56 & 0.64 \\ 0.64 & 0.16 \end{bmatrix}$$

3.0.9 Summing the Covariance Matrices

The total covariance matrix for Class 2 is:

$$S_{C2} = \frac{1}{n} \sum_{k=1}^n S_k$$

$$S_{C2} = \frac{1}{5} \left(\begin{bmatrix} 0.36 & -1.44 \\ -1.44 & 5.74 \end{bmatrix} + \begin{bmatrix} 5.76 & -0.96 \\ -0.96 & 0.16 \end{bmatrix} + \begin{bmatrix} 0.36 & -1.56 \\ -1.56 & 6.76 \end{bmatrix} + \begin{bmatrix} 0.16 & 0.24 \\ 0.24 & 0.36 \end{bmatrix} + \begin{bmatrix} 2.56 & 0.64 \\ 0.64 & 0.16 \end{bmatrix} \right)$$

$$S_{C2} = \begin{bmatrix} 1.84 & -0.04 \\ -0.04 & 2.64 \end{bmatrix}$$

Within-Class Scatter Matrix (S_W)

Adding S_{c1} and S_{c2} will give us S_W , the **Within-Class Scatter Matrix**:

$$S_W = \begin{bmatrix} 0.8 & -0.4 \\ -0.4 & 2.6 \end{bmatrix} + \begin{bmatrix} 1.84 & -0.04 \\ -0.04 & 2.64 \end{bmatrix} = \begin{bmatrix} 2.64 & -0.44 \\ -0.44 & 5.28 \end{bmatrix}$$

Step 2: Between-Class Scatter Matrix (S_B)

The **Between-Class Scatter Matrix** is given by:

$$S_B = (\mu_{c1} - \mu_{c2})(\mu_{c1} - \mu_{c2})^T$$

Where:

$$\begin{aligned} \mu_{c1} &= [3 \quad 3.6], \quad \mu_{c2} = [8.4 \quad 7.6] \\ \mu_{c1} - \mu_{c2} &= [3 \quad 3.6] - [8.4 \quad 7.6] = [-5.4 \quad -4] \\ S_B &= \begin{bmatrix} -5.4 \\ -4 \end{bmatrix} [-5.4 \quad -4] = \begin{bmatrix} 29.16 & 21.6 \\ 21.6 & 16 \end{bmatrix} \end{aligned}$$

Step 3: Find the Best LDA Projection Vector

We find the projection vector using the eigenvector with the largest eigenvalue. The eigenvector equation is:

Eigenvalue Calculation for $S_w^{-1}S_B$

Given Matrices

The within-class scatter matrix S_w and between-class scatter matrix S_B are:

$$S_w = \begin{bmatrix} 2.64 & -0.44 \\ -0.44 & 5.28 \end{bmatrix}, \quad S_B = \begin{bmatrix} 29.16 & 21.6 \\ 21.6 & 16 \end{bmatrix}.$$

We calculate $S_w^{-1}S_B$ step by step.

Step 1: Inverse of S_w

The inverse of S_w is given by:

$$S_w^{-1} = \frac{1}{\det(S_w)} \begin{bmatrix} 5.28 & 0.44 \\ 0.44 & 2.64 \end{bmatrix}.$$

The determinant of S_w is:

$$\det(S_w) = (2.64)(5.28) - (-0.44)(-0.44) = 13.9392 - 0.1936 = 13.7456.$$

Thus:

$$S_w^{-1} = \frac{1}{13.7456} \begin{bmatrix} 5.28 & 0.44 \\ 0.44 & 2.64 \end{bmatrix}.$$

Step 2: Compute $S_w^{-1}S_B$

Now multiply S_w^{-1} with S_B :

$$S_w^{-1}S_B = \frac{1}{13.7456} \begin{bmatrix} 5.28 & 0.44 \\ 0.44 & 2.64 \end{bmatrix} \begin{bmatrix} 29.16 & 21.6 \\ 21.6 & 16 \end{bmatrix}.$$

Simplifying the product, we get:

$$S_w^{-1}S_B = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix},$$

where numerical values for $\lambda_{11}, \lambda_{12}, \lambda_{21}, \lambda_{22}$ are substituted.

Step 3: Eigenvalue Equation

The characteristic equation is defined as:

$$\det(S_w^{-1}S_B - \lambda I) = 0.$$

Expanding the determinant:

$$\det \begin{bmatrix} \lambda_{11} - \lambda & \lambda_{12} \\ \lambda_{21} & \lambda_{22} - \lambda \end{bmatrix} = (\lambda_{11} - \lambda)(\lambda_{22} - \lambda) - \lambda_{12}\lambda_{21}.$$

This simplifies to a quadratic equation:

$$\lambda^2 - \text{Tr}(S_w^{-1}S_B)\lambda + \det(S_w^{-1}S_B) = 0,$$

where Tr is the trace of the matrix.

Step 4: Solving the Quadratic Equation

Substitute the trace $\text{Tr}(S_w^{-1}S_B)$ and determinant:

$$\lambda^2 - 15.65\lambda + 0 = 0.$$

The roots of this equation are:

$$\lambda_1 = 15.65, \quad \lambda_2 = 0.$$

Step 5: Eigenvector Calculation for $\lambda = 15.65$

We start with the eigenvalue equation:

$$(S_w^{-1}S_B - \lambda I) \mathbf{v} = 0,$$

where:

$$S_w^{-1}S_B = \frac{1}{13.7456} \begin{bmatrix} 5.28 & 0.44 \\ 0.44 & 2.64 \end{bmatrix} \begin{bmatrix} 29.16 & 21.6 \\ 21.6 & 16 \end{bmatrix} = \begin{bmatrix} 11.52 & 8.4 \\ 8.4 & 6.12 \end{bmatrix}.$$

Subtract λI from $S_w^{-1}S_B$:

$$\begin{bmatrix} 11.52 & 8.4 \\ 8.4 & 6.12 \end{bmatrix} - 15.65 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -4.13 & 8.4 \\ 8.4 & -9.53 \end{bmatrix}.$$

The eigenvalue equation becomes:

$$\begin{bmatrix} -4.13 & 8.4 \\ 8.4 & -9.53 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0.$$

Solve for v

Expanding the system of equations:

$$-4.13v_1 + 8.4v_2 = 0,$$

$$8.4v_1 - 9.53v_2 = 0.$$

From the first equation, solve for v_1 in terms of v_2 :

$$v_1 = \frac{8.4}{4.13}v_2 \approx 2.034v_2.$$

Normalize $v_2 = 1$:

$$v_1 \approx 2.34.$$

Thus, the eigenvector is:

$$\mathbf{v} = \begin{bmatrix} 2.34 \\ 1 \end{bmatrix}.$$

Step 4: Transformation onto New Subspace

$$Y = XW, \quad W = \begin{bmatrix} 2.34 \\ 1 \end{bmatrix}$$

Original dataset X :

$$\begin{bmatrix} 4 & 1 \\ 2 & 4 \\ 2 & 3 \\ 3 & 6 \\ 4 & 4 \\ 9 & 10 \\ 6 & 8 \\ 9 & 5 \\ 8 & 7 \\ 10 & 8 \end{bmatrix}$$

Transformed dataset Y :

$$Y = X \times \begin{bmatrix} 2.34 \\ 1 \end{bmatrix} = \begin{bmatrix} 10.36 \\ 8.68 \\ 7.68 \\ 13.02 \\ 13.36 \\ 31.36 \\ 22.04 \\ 26.06 \\ 25.72 \\ 31.4 \end{bmatrix}$$

Transformation Illustration

X1	X2		X
4	1		10.36
2	4		8.68
2	3		7.68
3	6		13.02
4	4		13.36
9	10		31.36
6	8		22.04
9	5		26.06
8	7		25.72
10	8		31.4

Transformation

Figure 4: Transformation from X_1, X_2 to X

4 Methodology

4.1 Data Collection

The dataset utilized in this study was sourced from publicly available Human Development Index (HDI) records, which include various socio-economic indicators such as Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per capita. The dataset contains 193 observations representing different countries, along with their categorical classifications (e.g., “Very High Developed”).

4.2 Data Preprocessing

Before applying clustering algorithms, the dataset was cleaned and preprocessed to ensure consistency and reliability of the analysis. Specifically:

1. **Numeric Feature Selection:** Only numeric variables were selected for clustering analysis to avoid issues with categorical data. These features included Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per capita.
2. **Scaling:** The numeric data was scaled using standardization to ensure that all variables had equal weight in the clustering process. This step mitigates the effect of differing units and magnitudes across features.
3. **Feature Exclusion:** The calculated HDI value was excluded from the clustering process to allow the algorithm to identify patterns independently of the predefined HDI score.

4.3 Clustering Approach

To evaluate the alignment between HDI categories and clustering results, K-means clustering was applied to the preprocessed dataset. The methodology involved the following steps:

1. **Elbow Method:** To determine the optimal number of clusters, the Elbow Method was used, which evaluates the within-cluster sum of squares (WSS) for various values of K. Based on the results, K=3 and K=4 were selected for further analysis.
2. **Implementation:** The K-means algorithm was executed for both K=3 and K=4 using a fixed random seed to ensure reproducibility. The clusters were then appended to the dataset for interpretation.
3. **Cluster Labeling:** Cluster labels were assigned to each observation, and a comparison was made between these labels and the predefined HDI categories.

4.4 Validation with Linear Discriminant Analysis (LDA)

To assess the discriminative power of the clusters and validate their alignment with HDI categories, Linear Discriminant Analysis (LDA) was performed:

1. **Prediction and Evaluation:** The model's predictions were compared to the original HDI categories, allowing for an assessment of how well the clustering captured the existing classification structure.

5 Discussion

The results from clustering analysis offer a valuable strategy for improving crisis management and intervention planning. By categorizing countries based on essential socio-economic indicators such as life expectancy, education levels, and income, this technique enables the identification of nations that might require immediate humanitarian aid in times of crisis. The groups formed, especially those with lower Human Development Index (HDI) values, often face heightened vulnerabilities such as underdeveloped healthcare systems and infrastructure, which make them more susceptible to the negative impacts of natural disasters, conflicts, and other crises.

In disaster relief efforts, this approach proves especially useful. After a disaster or during post-conflict recovery, identifying countries with the lowest HDI can guide humanitarian organizations in prioritizing their support. Furthermore, this data-driven approach ensures that resources are allocated effectively, reducing inefficiencies in aid distribution and boosting the overall success of relief operations.

Additionally, comparing these clusters to existing HDI classifications further reinforces the relevance of HDI in crisis contexts. The alignment between the clusters and HDI categories validates the reliability of HDI as a tool for classifying countries based on socio-economic factors, strengthening its role in crisis preparedness.

In conclusion, this study underscores the importance of clustering analysis as an effective method for enhancing crisis management. By leveraging this approach, humanitarian efforts can be more precise, responsive, and better equipped to direct resources to where they are most urgently needed during crises.

6 References

1. United Nations Development Programme. "Human Development Report 2023." UNDP, 2023. Available at: <https://hdr.undp.org/>
2. Zhao, S., Zhang, B., Yang, J., Zhou, J., & Xu, Y. (2024). Linear discriminant analysis. *Nature Reviews Methods Primers*.
3. Fang, E. X., Mei, Y., Shi, Y., Xu, Q., & Zhao, T. (2023). Pivotal estimation of linear discriminant analysis in high dimensions. *Journal of Machine Learning Research*.
4. Zhuang, Y., Chen, X., Yang, Y., & Zhang, R. Y. (2023). Statistically optimal K-means clustering via nonnegative low-rank semidefinite programming. *arXiv preprint arXiv:2305.18436*.
5. Layeb, A. (2023). CKmeans and FCKmeans: Two deterministic initialization procedures for Kmeans algorithm using a modified crowding distance.
6. Alkire, S., Foster, J. "Counting and Multidimensional Poverty Measurement." *Journal of Public Economics*, 2011.
7. Sachs, J. D., et al. "The Sustainable Development Goals and COVID-19." *Nature*, 2020.
8. Fisher, R. A. "The Use of Multiple Measurements in Taxonomic Problems." *Annals of Eugenics*, 1936.
9. Fisher, R. A. "The Use of Multiple Measurements in Taxonomic Problems". *Annals of Eugenics*, 1936.
10. McLachlan, G. J. "Discriminant Analysis and Statistical Pattern Recognition". Wiley, 2004.
11. Duda, R. O., Hart, P. E., and Stork, D. G. "Pattern Classification". Wiley-Interscience, 2001.
12. Bishop, C. M. "Pattern Recognition and Machine Learning". Springer, 2006.